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Influence of argon hydrogen plasma treatment on the wettability, void and high temperature reflow reliability of indium thermal interface material for FCBGA package



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Abstract

Indium thermal interface materials (TIMs) exhibit significant potential for flip-chip ball grid array (FCBGA) packaging due to their high thermal conductivity (82W/(m·K)) and remarkable ductility. However, the ultra-thin gold layer often fails to adequately protect against nickel oxidation, which compromises the wettability of the indium TIM and results in the formation of a non-uniform interfacial intermetallic compound (IMC), thereby hindering broader adoption. This study introduces an Ar+H₂ plasma

surface treatment aimed at enhancing the wettability of indium TIM on backside metallization (BSM) chips and promoting uniform IMC formation. Experimental results indicate that, following treatment, the indium TIM achieves a spreading ratio of 112.7% on BSM chips after nine high-temperature reflows, reflecting a 25.7% improvement compared to untreated samples. Furthermore, a uniform $(\text{Ni}, \text{Au})_{28}\text{In}_{72}$ layer develops at the interface, significantly improving wetting performance after multiple reflows. In the Lid/In/BSM chip sandwich structure, the void size percentage for Ar+H₂ plasma-treated samples is only 1.1% after nine reflows, which is 39% lower than that of untreated samples. Thermal simulation results reveal that the Ar+H₂ plasma-treated sample, exhibiting the lowest void ratio, attains optimal thermal performance. The developed process is compatible with existing packaging production lines, providing a feasible technical pathway for the industrial application of indium TIMs in FCBGAs packaging.



Keywords

—Indium thermal interface materials; Plasma surface treatment; Intermetallic compounds; Spreading ratio; Void size percentage

1. Introduction

As modern electronic devices progress toward greater integration and power density, effective thermal management becomes essential for ensuring device reliability and performance [1], [2], [3]. Among the various thermal interface materials (TIMs), indium is notable for its outstanding thermal conductivity (82 W/(m·K)) and its capacity for plastic deformation [4]. However, indium TIMs face significant challenges in practical packaging applications. During processing, gas entrapment and insufficient solder spreading frequently occur, resulting in an increased void size percentage that negatively affects heat conduction efficiency and interfacial mechanical strength [5,6]. Furthermore, brittle intermetallic compounds (IMCs) readily form between indium and metal heat dissipation components, increasing the likelihood of interface failure [7,8]. Therefore, minimizing the void size percentage and regulating the uniform growth of IMCs are critical for promoting the industrialization of indium thermal interface materials.

To achieve effective bonding between indium thermal interface materials and silicon chips, it is essential to subject the backside of the chips to a metallization treatment. The commonly employed Ti/Ni/Au multi-layer structure serves specific functions: the Ti layer enhances adhesion, the Ni layer functions as a diffusion barrier, and the Au layer improves wetting performance. Research suggests that an optimal thickness for the gold layer is critical. A gold layer that is too thin fails to adequately prevent Ni diffusion and does not maintain satisfactory wetting. In contrast, a thicker gold layer improves initial wetting but accelerates the formation of brittle intermetallic compounds (IMCs) and increases manufacturing costs [9,10]. Furthermore, simply increasing the thickness of the gold layer does not effectively mitigate interface oxidation. This challenge has prompted researchers to explore more practical engineering solutions.

The morphological evolution of the interfacial intermetallic compound (IMC) is a critical factor influencing wettability. Research indicates that during the reactive wetting process, a rough IMC surface can significantly impair wetting performance [11]. This phenomenon is primarily due to the pinning effect, in which the rough IMC surface anchors the solid-liquid-gas three-phase contact line, thereby increasing the energy barrier for the advancement of the wetting front [12]. As a result, this leads to an elevated contact angle and a diminished solder spreading area. Therefore, achieving a smoother interfacial morphology through the control of the IMC formation process has emerged as a vital strategy for enhancing wettability [13]. Despite numerous studies investigating the effects of IMC roughness on reliability, research focused on methods to flatten IMC morphology remains relatively scarce [[14], [15], [16], [17]].

Plasma surface treatment technology offers an innovative solution to these challenges. This approach markedly enhances the wetting performance of solder by effectively eliminating surface contaminants and oxide layers [[18], [19], [20], [21], [22]]. J.K. Moon et al. demonstrated that argon-hydrogen plasma treatment can facilitate flux-less soldering; however, the stringent environmental control requirements of the process restrict its industrial applicability [23]. L. Vivet et al. observed that high-power plasma treatment may lead to surface reoxidation issues [24]. Importantly, research conducted by Sagung Dewi Kencana indicates that the atmospheric plasma process yields a more uniform oxide layer, thereby significantly improving the solder spreading ratio [25].

Fig. 1 illustrates that in engineering practice, the BSM chip and indium thermal interface material undergo several heat treatment steps, including high-temperature reflow and baking, prior to welding. Studies indicate that metal materials are prone to surface oxidation and contamination during heat treatment in an atmospheric environment [26]. Although the Au layer coating on the BSM chip acts as a protective barrier, it is challenging to entirely prevent the deterioration of interface wettability due to

oxidation. This study focuses on the heat-treated BSM chip and designs two plasma treatment processes with different atmospheres, utilizing untreated samples as a control. The objective is to systematically investigate how plasma treatment influences the surface characteristics of the coating, the spreading ratio, and the void size percentage of the samples. Specifically, the research entails: (1) analyzing the evolution of the surface chemical state and morphology of the coating under various treatment conditions; (2) assessing the spreading performance of metallic indium on the BSM chip surface after multiple high-temperature reflows; and (3) measuring the void size percentage in the Lid/In/BSM sandwich structure following repeated high-temperature reflows. The findings provide valuable insights for the engineering application of indium TIM in FCBGA packaging.



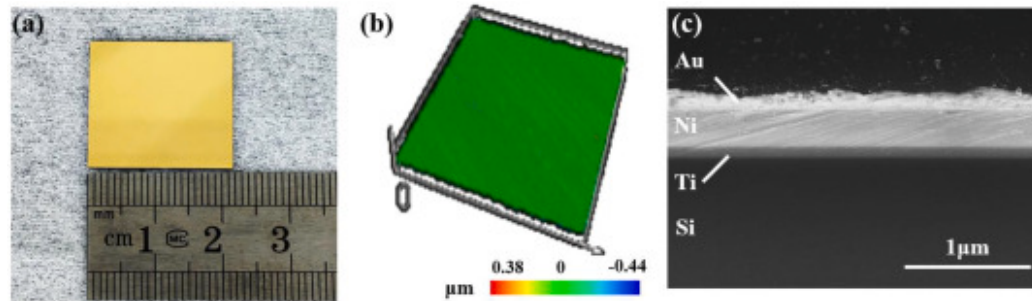
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Fig. 1. Process flow of BSM chip before welding with indium thermal interface material.

2. Research methods and experimental procedures

[Fig. 2\(a\)](#) presents the macroscopic morphology of the heat-treated BSM chip. The surface roughness, assessed using a 3D laser confocal microscope (OLS500, Olympus), is illustrated in [Fig. 2\(b\)](#), indicating an average surface roughness R_a of less than $0.01\ \mu\text{m}$. [Fig. 2\(c\)](#) displays the cross-sectional microstructure of the BSM chip. A $0.1\text{-}\mu\text{m}$ Ti adhesion layer, a $0.3\text{-}\mu\text{m}$ Ni barrier layer, and a $0.1\text{-}\mu\text{m}$ Au wetting layer were sequentially deposited on the silicon chip surface through vacuum evaporation deposition.

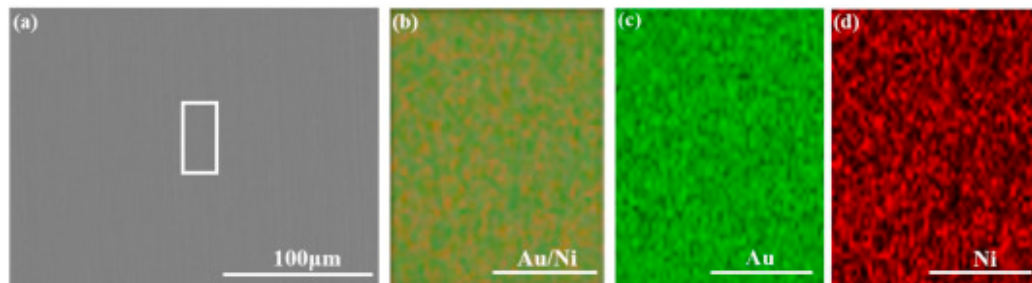


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Fig. 2. Illustrates the characteristics of the BSM chip following heat treatment: (a) the macroscopic surface morphology, (b) the three-dimensional surface morphology, and (c) the SEM image of the cross-section.

Fig. 3 illustrates the elemental distribution characteristics on the coating surface of the BSM chip. In Fig. 3(b)–(d), only Au and Ni elements are detected on the coating surface, with no signals from Ti and Si elements. This absence indicates effective adhesion of the Ti layer and a uniform, dense structure of the Ni layer. Notably, the detection of Ni signals in the Au layer region suggests that the Au layer does not completely cover the Ni layer.



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Fig. 3. Element distribution of BSM chip coating after heat treatment.

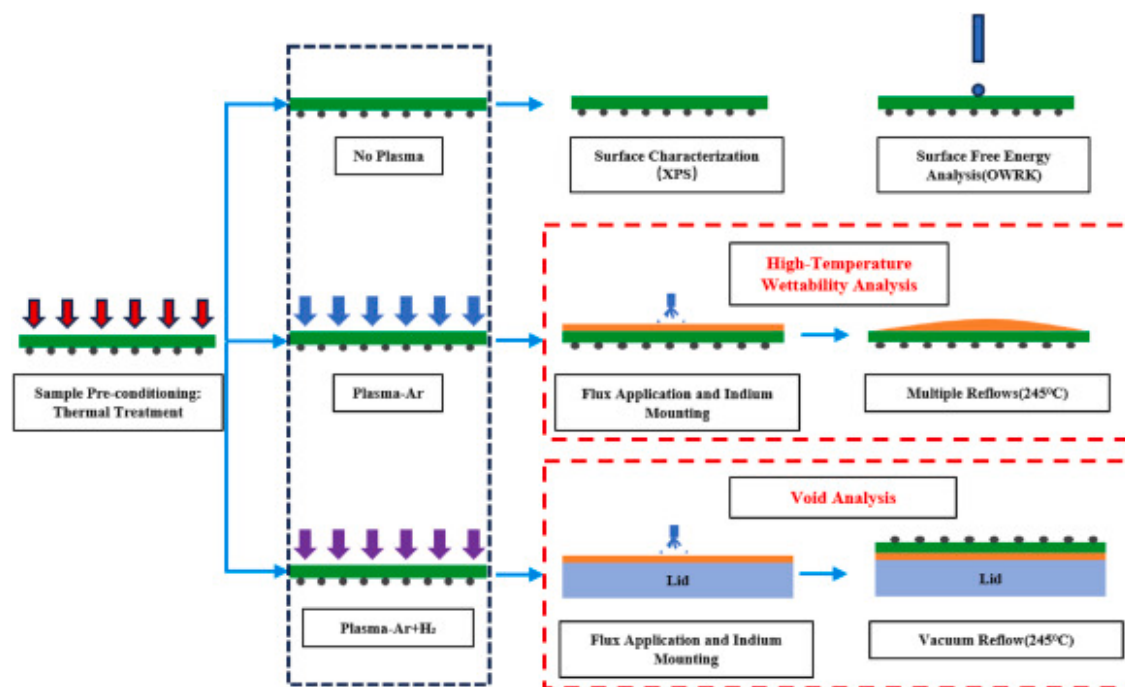
In this study, we investigated heat-treated BSM chips through three comparative experiments: a reference group (No Plasma), a pure argon plasma treatment group (Plasma-Ar), and an argon-hydrogen mixed gas plasma treatment group (Plasma-Ar+H₂). The samples were processed using a cavity-type plasma cleaning system (VSP-88H, VISION semicon). The set and actual operating parameters of the equipment are summarized in Table 1. All experiments were conducted at room temperature; however, it is important to note that the sample temperature was not monitored in real time during the plasma treatment. Following plasma processing, the samples were promptly stored in a temperature- and humidity-controlled nitrogen cabinet (22.4°C, 37.3% RH) to facilitate a controlled re-oxidation process. The time interval between the completion of plasma treatment and subsequent high-temperature reflow, as well as voiding tests, was strictly limited to 4h.

Table 1. The parameters of plasma process.

Items	Set Parameters	Actual Parameter
Gas type-1	100Ar	100Ar
Gas type-2	95Ar+5H ₂	95Ar+5H ₂
Chamber pressure(mTorr)	0	81
Power(W)	550	549
Flux-1(sccm)	20	20.3(Ar,100%)
Flux-2(sccm)	20	19.3(Ar,95%); 1.0(H ₂ ,5%)
Time(s)	800	800
Power density(W/cm ²)	0.75	0.75
Electrode distance(mm)	146	146

To systematically evaluate the effects of surface treatment, this study delineates three critical testing steps: analyzing the coating surface state, assessing high-temperature wettability, and measuring the void size percentage of the sample. Fig. 4 illustrates the overall process. The materials utilized in this study include the lid supplied by Suzhou SITRI Material Tech Co., Ltd. (SITRI, China),

the BSM chip fabricated by Jiangsu CAS Microelectronics Integration Technology Co., Ltd. (CASMEIT, China), an indium sheet (purity 99.99%) produced by Indium Corporation (Indium, America), and flux from Senju Metal Industry (Senju, Japan). To mitigate the influence of flux on the experimental outcomes, both water-soluble (WS) and no-clean (NC) flux types were selected for comparison in the high-temperature wettability and void fraction tests. The properties of the various fluxes are summarized in Table S1. In the high-temperature wettability experiment, the indium sheet was laser-micro processed into rectangles measuring $5 \times 4 \times 0.23$ mm to prevent overflow of the molten indium. This sheet was then mounted on a 22×21 mm BSM chip. For the void fraction tests, a $22 \times 21 \times 0.23$ mm indium sheet was utilized, with dimensions corresponding to those of the BSM chip. The length and width of the indium sheets were measured using a Vernier caliper, while thickness was determined with a micrometer, as illustrated in Fig. S1. Statistical analysis revealed that the ranges for length, width, and thickness were 0.032 mm, 0.066 mm, and 0.004 mm, respectively, with standard deviations of 0.0105 mm, 0.0214 mm, and 0.0011 mm. These findings indicate good machining consistency and confirm that the indium sheets satisfy the requirements for subsequent bonding processes.



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Fig. 4. Analysis flow of samples treated with different plasma.

The chemical states, oxide concentration, and carbon bonding structure of the BSM chip coating surface were analyzed using X-ray photoelectron spectroscopy (XPS, Thermo Scientific K-Alpha). Following preparation, the samples were vacuum-sealed and briefly stored (<4h) in a nitrogen cabinet prior to immediate XPS measurement, ensuring that the acquired spectra accurately represented the surface chemistry. The obtained spectra were processed with Advantage software, employing the Shirley method for background subtraction and referencing charge correction to the C 1s peak of surface-adventitious carbon at 284.8eV. The Scofield sensitivity factors utilized were as follows: Au 4f=20.735, C 1s=1.000, Ni 2p=20.765, O 1s=2.881. Atomic percentages were calculated by normalizing the peak areas with the corresponding atomic sensitivity factors [27,28]. The fitting residual of the Ni 2p spectrum for the No plasma sample is depicted in Fig. S2.

The wetting behavior of the coating surface on the BSM chip was assessed using an optical contact angle goniometer (SDC-100SH, YWZR). Deionized water and ethylene glycol were employed as test liquids [29]. A 1 μ L micro syringe was utilized to deposit droplets onto the coating surface. The equilibrium contact angle was determined statically through the application of the Young-Laplace equation [30]. Surface free energy, including its disperse and polar components, was calculated using the OWRK (Owens-Wendt-Rabel-Kaelble) model [31]. Each set of conditions was tested five times, and the average value was recorded.

Both the high-temperature wettability and sample void fraction tests were performed in a vacuum reflow oven, with the oxygen content in the chamber maintained below 50ppm. The test samples were stored in a nitrogen cabinet for a brief period of less than 4h before undergoing the respective tests. The procedure for the high-temperature wettability test was as follows: first, flux was uniformly applied to the surface of the BSM chip, followed by the high-precision placement of the indium sheet using a chip mounter. An equal amount of flux was then sprayed onto the surface of the indium sheet. Reflow was subsequently conducted at a peak temperature of 245°C, with the reflow profile illustrated in Fig. S3. The spreading ratio was defined as the ratio of the spread solder area after reflow to the initial area, with the average value derived from five repeated tests per group. The post-spreading area was measured using an ultra-depth-of-field microscope (VHX-7000, KEYENCE) and its built-in software, as depicted in Fig. S4. The software automatically identified the solder region through threshold segmentation and calculated the pixel-based area, which was then converted into the actual physical area via system calibration. The specific imaging conditions were as follows: ZS20×objective lens, 50×magnification, ring-shaped incident light source, and a tilt angle set to 0°.

The procedure for assessing the void size percentage of samples commences with the uniform application of flux on the surface of the gold-plated lid. Subsequently, an indium sheet is accurately positioned on the flux coating, followed by an additional application of flux onto the surface of the indium sheet. The BSM chip is then affixed on top. Reflow soldering is conducted at a peak temperature of 245°C. The amount of flux is meticulously regulated between 1 and 1.5 mg, utilizing a high-precision electronic balance with a resolution of 0.01 mg. An X-ray device (Quadra7, Nordson) is then employed for the quantitative analysis of void size percentage. The detection area, which corresponds to the dimensions of the BSM chip, measures 22×21 mm. The maximum void size percentage is calculated by considering the size of the largest single solder void relative to the die size. The average void size percentage is determined by calculating the total void area in relation to the detection area. Each condition is evaluated using five samples to ensure reliability. The key X-ray parameters are as follows: power of 10W, tube voltage of 120kV, and image frame averaging configured for 128 frames (128 Avg). Void identification and calculation are conducted using the equipment's built-in software, as illustrated in Fig. S8. The "Blob" analysis algorithm is employed, with the maximum void fraction parameter set to 50%, the contrast multiplier adjusted to 15, and the edge exclusion fixed at 0%. The measurement field of view (FOV) is established at 48.495 mm.

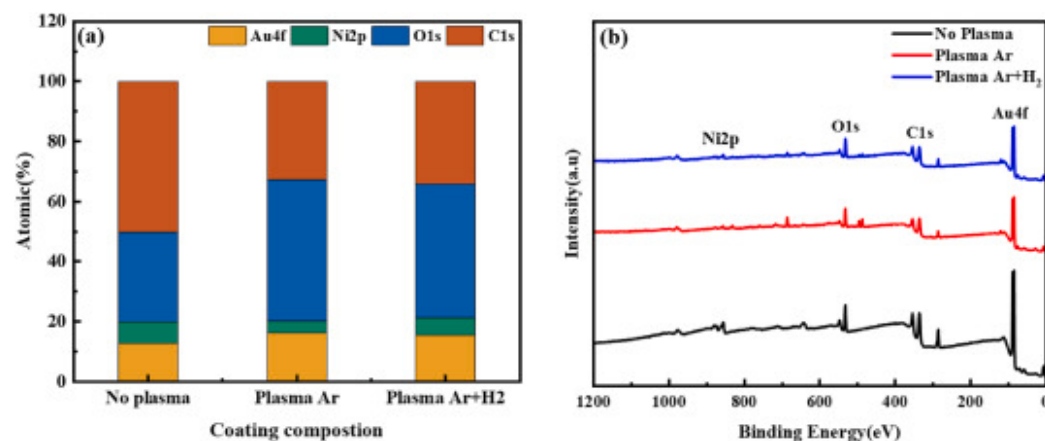
The morphology and interfacial microstructure of the intermetallic compound (IMC) were analyzed using a scanning electron microscope (SEM, FEI Quanta200F). The composition of the IMC at the sample interface was determined through energy dispersive spectroscopy (EDS). To examine the IMC morphology, a mixed solution consisting of 13% HNO₃, 10% HCl, and 77% CH₃OH was utilized to etch the indium layer, thereby revealing the IMC layer formed at the interface between the indium and the BSM chip plating. For measuring the IMC thickness, the sample was first cured with epoxy resin, followed by preparation of the analysis cross-section using a diamond wire cutting technique. The cross-section was then sequentially ground with 180#, 1200#, 2400#, and 4000# silicon carbide sandpapers and polished with an Al₂O₃ and SiO₂ suspension. The average thickness of the IMC was calculated by dividing the total area of the observed region by its corresponding length. All statistical results were derived from average values measured across at least five fields of view.

3. Results and discussion

3.1. Surface chemical state evolution

Fig. 5 depicts the variations in elemental composition and X-ray photoelectron spectroscopy (XPS) survey spectra results across different coating treatment methods. In the sample lacking plasma treatment, in addition to the predominant element Au, there

are prominent signals for O1s (29.74%) and C1s (50.24%), as well as a substantial signal for Ni2p (7.34%). This observation is consistent with prior research, which indicates that during process transfer, even if the Au protective layer remains intact, Ni atoms can migrate to the surface through grain boundary diffusion, subsequently forming nickel oxide in the presence of air [10]. The elevated C1s proportion (50.24%) in the untreated sample suggests considerable organic contamination on the surface, while the O1s proportion (29.74%) primarily derives from the surface oxide layer. After plasma treatment, the C1s content decreased significantly to 32.81–33.97%, indicating that plasma in both atmospheres effectively eliminates surface carbon contamination. Following the Plasma-Ar treatment, the O1s proportion rose to 47%, likely due to the exposure of the oxidized Ni layer after the removal of organic contaminants or as a result of further oxidation during the treatment. Under the Plasma-Ar+H₂ treatment conditions, the O1s proportion slightly declined to 44.76%, suggesting that the reducibility of H₂ somewhat inhibited the oxidation process, although oxides remained predominant overall. In summary, plasma treatment not only effectively cleans the coating surface but also significantly alters its surface chemical state. Furthermore, the introduction of H₂ moderately regulates the extent of surface oxidation.



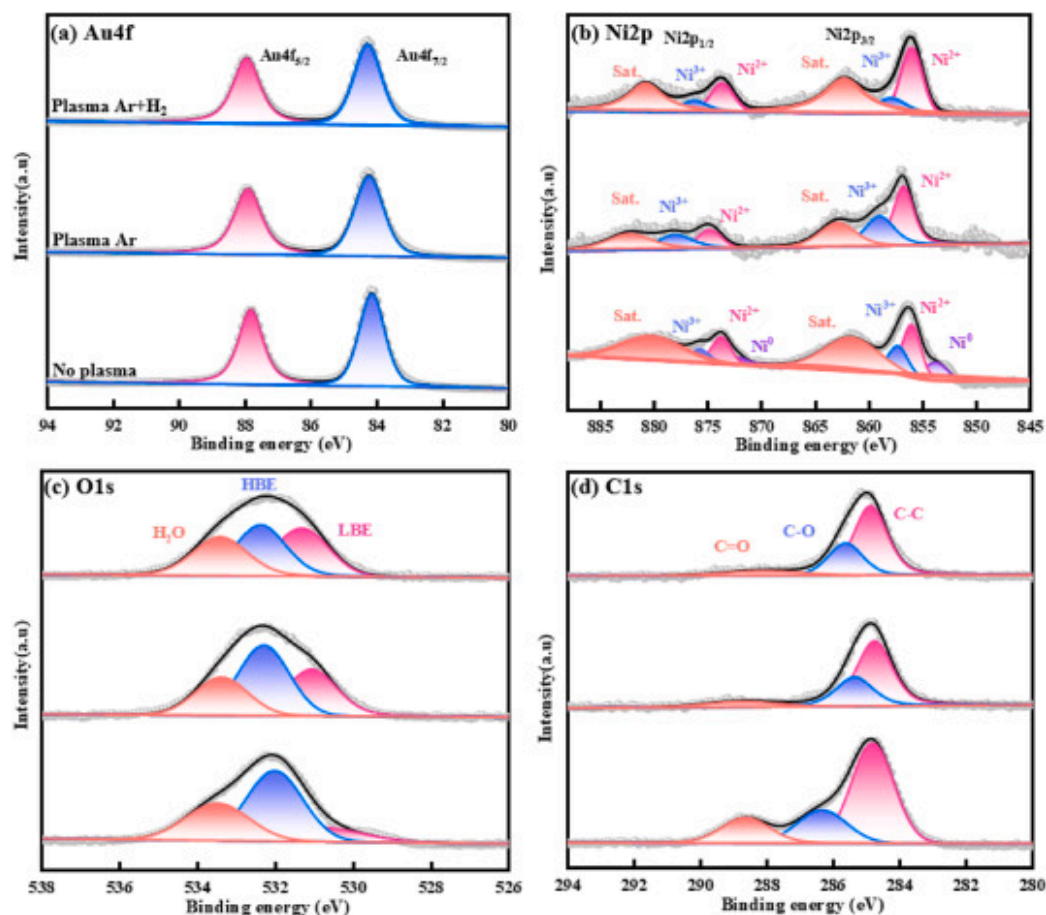
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Fig. 5. Elemental composition ratios and XPS survey spectra for different surface treatments.

Fig. 6(a)–(d) depict the variations in surface chemical states of samples subjected to different treatment methods. In Fig. 6(a), the No plasma sample displays characteristic Au 4f peaks at 84.15eV and 87.94eV, signifying the presence of metallic Au⁰ [32,33]. In

contrast, the samples treated with Plasma-Ar and Plasma-Ar+H₂ exhibit no significant chemical shifts. This observation suggests that the Au layer preserves its metallic state throughout the treatment process, thereby maintaining stable chemical properties.



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Fig. 6. XPS spectrum of different surface treatments (a)Au4f spectrum(b)Ni2p spectrum(c)O1s spectrum(d)C1s spectrum.

The evolution of the chemical state of Ni 2p is depicted in Fig. 6(b). In the untreated sample, characteristic peaks of metallic Ni were observed at 852.7 eV and 870.0 eV [34,35]. Peaks at 856.0 eV and 873.3 eV were assigned to Ni²⁺ in the Ni 2p_{3/2} and Ni 2p_{1/2} orbitals [36], respectively, while peaks at 857.0 eV and 876.3 eV corresponded to Ni³⁺ in the same orbitals [37,38]. Furthermore,

satellite peaks at 861.2 eV and 879.1 eV confirmed the presence of Ni^{2+} , which typically exists as NiO or $\text{Ni}(\text{OH})_2$ [35]. These findings suggest that during the preceding heat treatment, Ni atoms infiltrated the thin Au layer and underwent oxidation. After Plasma-Ar treatment, the intensity of the metallic Ni peaks diminished, while the Ni^{3+} peaks exhibited a significant increase, indicating enhanced oxidation on the sample surface. High-energy Ar ions excited gaseous oxygen and water molecules, resulting in the generation of numerous reactive oxygen radicals ($\cdot\text{O}$, $\cdot\text{OH}$). These radicals oxidized $\text{Ni}(\text{OH})_2$ and Ni directly, resulting in the formation of Ni^{2+} vacancies and Ni^{3+} (which exists as NiOOH or Ni_2O_3) through a non-stoichiometric oxidation process characterized by hole doping due to an excess of oxygen [38,39]. The calculated $\text{Ni}^{3+}/\text{Ni}^{2+}$ ratios, derived from peak areas, are 0.62 for the No plasma sample, 0.73 for the Plasma-Ar sample, and 0.43 for the Plasma-Ar+ H_2 sample. These findings clearly demonstrate that treatment with Plasma-Ar+ H_2 significantly diminishes the characteristic peak of Ni^{3+} while enhancing that of Ni^{2+} . This suggests that Ar+ H_2 plasma can partially reduce Ni^{3+} to Ni^{2+} .

Fig. 6(c) demonstrates that the O1s spectrum indicates alterations in the oxidation state of metallic Ni. The peaks observed at binding energies of 530.1 eV and 531.1 eV correspond to low and high binding energies, respectively [39]. Research has established that the coexistence of Ni^{2+} and Ni^{3+} results in oxygen defects, which enhance interactions between reactants and intermediates such as OH^- , O, OH, and OOH^* [40,41]. Additionally, the O1s spectrum displays a peak at 532.8 eV, which is attributed to water from natural contamination due to exposure to ambient conditions [10]. Fig. 6(d) illustrates the evolution of C1s chemical states. In samples lacking plasma treatment (No plasma), C–C bonds (284.5 eV), C–O bonds (286.6 eV), and C=O bonds (288.8 eV) are predominant. After plasma treatment, the intensity of the C–C peak decreases significantly [42,43].

Table II indicates that X-ray photoelectron spectroscopy (XPS) quantitative analysis demonstrates that plasma treatment significantly affects the surface chemical states of the material. In the untreated sample, the contents of Ni^{2+} and Ni^{3+} are 3.60 at % and 2.23 at %, respectively. After Plasma-Ar treatment, the Ni^{3+} content increases to 3.73 at %, while the Ni^{2+} content decreases to 2.36 at %, suggesting that Plasma-Ar treatment promotes the formation of high-valence nickel oxides. Notably, following Plasma-Ar+ H_2 treatment, the Ni^{3+} content declines to 1.70 at %, illustrating the reduction effect of H^+ . Regarding the evolution of carbon species, the untreated sample exhibits a high proportion of C–C bonds at 29.76 at %. Following Ar plasma and Ar+ H_2 plasma treatments, the C–C bond contents decrease to 20.00 at % and 21.48 at %, respectively, confirming that plasma treatments in different atmospheres effectively eliminate organic contaminants from the surface.

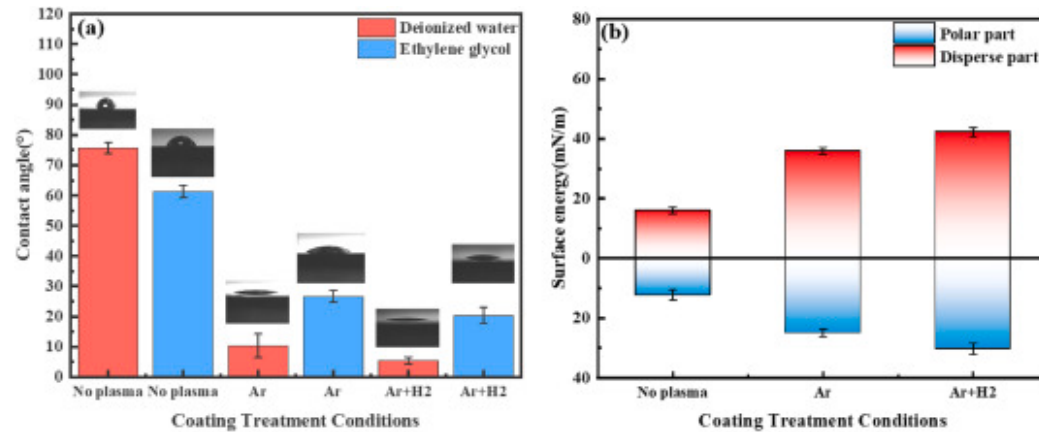
Table II. Atomic ratio distribution of Ni and C surface chemical states under different plasma treatments.

Treatment	Ni ⁰ (at%)	Ni ²⁺ (at%)	Ni ³⁺ (at%)	C-C(at%)	C-O(at%)	C=O(at%)
No plasma	1.50517	3.60384	2.23099	29.7605	11.4062	9.07329
Plasma-Ar	0	2.36395	3.72605	20.0038	9.47383	3.3324
Plasma-Ar+H ₂	0	3.394378	1.69622	21.4795	9.72443	2.76606

Note: The data are calculated based on the peak fitting results.

3.2. Surface energy and morphology evolution

Surface energy plays a critical role in characterizing solid surfaces and is typically categorized into disperse and polar components. Its magnitude significantly affects a material's wettability. This study systematically assessed the wetting behavior of the coating surface under various treatment conditions through contact angle measurements. As shown in Fig. 7(a), the No Plasma sample demonstrated weak hydrophilicity, with contact angles of $75.7^\circ \pm 1.75^\circ$ for deionized water and $61.33^\circ \pm 2.01^\circ$ for ethylene glycol. The Plasma-Ar treatment markedly enhanced wettability, resulting in a reduction of the water contact angle to $10.31^\circ \pm 3.88^\circ$ and the ethylene glycol angle to $26.59^\circ \pm 1.97^\circ$. After the Plasma-Ar+H₂ treatment, the coating surface exhibited super hydrophilicity, with the water contact angle further decreasing to $5.47^\circ \pm 1.12^\circ$ and the ethylene glycol angle to $20.31^\circ \pm 2.66^\circ$. These results indicate that plasma treatment, particularly Plasma-Ar+H₂, effectively improves wetting performance.



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Fig. 7. Surface properties of different surface treatments. (a)contact angle(b)surface energy.

The OWRK model, a well-established framework based on the two-component theory of surface energy, decomposes the solid surface free energy into disperse and polar components [31]. By employing the contact angles of two liquids with known surface tension properties on the solid surface, the model can concurrently ascertain both the disperse and polar components. It then computes the total surface energy. The fundamental relationship is expressed as follows:

$$\gamma_L (1 + \cos \theta) = 2 \left(\sqrt{\gamma_S^d \gamma_L^d} + \sqrt{\gamma_S^p \gamma_L^p} \right) \quad (1-1)$$

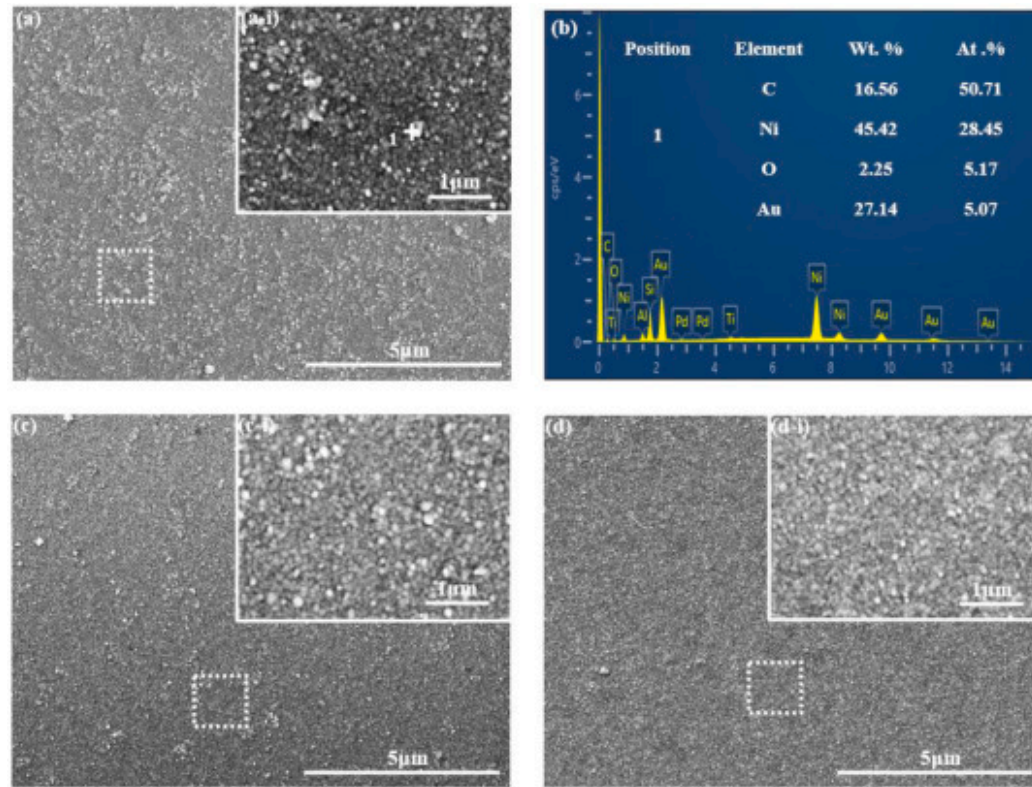
In this context, γ_L denotes the liquid surface tension, while γ_L^d and γ_L^p represent the disperse and polar components of the liquid, respectively. Similarly, γ_S^d and γ_S^p are the disperse and polar components of the solid. The contact angle of the liquid on the solid surface is indicated by θ . Table III presents the polarities of the liquids utilized in this study's static drop test. By analyzing the measurement results from two liquids, the values of γ_S^d and γ_S^p for the sample surface can be determined, allowing for the calculation of its total surface energy.

Table III. Polar properties of the liquids used for contact angle measurement for the sessile drop test.

	Surface tension γ_L (mN/m)	Disperse part γ_L^d (mN/m)	Polar part γ_L^p (mN/m)
Deionized water	72.8	26	46.8
Ethylene glycol	47.7	26.4	21.3

The OWRK model calculations further quantified the impact of plasma treatment on surface energy regulation. As shown in Fig. 7(b), the No Plasma sample displayed a total surface energy of 28.25 mN/m, consisting of a disperse component of 16.00 mN/m and a polar component of 12.25 mN/m, which is characteristic of low surface energy. After Plasma-Ar treatment, the total surface energy increased to 61.00 mN/m, with the disperse component rising to 36.00 mN/m and the polar component to 25.00 mN/m. This treatment not only enhanced non-polar interactions but also significantly elevated the density of surface polar functional groups. Notably, the Plasma-Ar+H₂ treatment resulted in an even greater enhancement in surface energy, achieving a total of 72.50 mN/m, which is approximately 157% higher than that of the No Plasma sample. The disperse and polar components increased to 42.25 mN/m and 30.25 mN/m, respectively. This synergistic enhancement indicates that the introduction of H₂ optimized surface cleanliness through the reduction effect of hydrogen radicals and facilitated a more uniform distribution of polar functional groups (such as -OH, -O-), thereby comprehensively improving the surface energy characteristics.

The surface morphologies of BSM chip coatings were characterized after various surface treatments, and the surface components of the No plasma sample were analyzed. In Fig. 8(a), numerous bright white particles are evident on the surface of the No plasma sample, along with several stacked spherical structures. EDS analysis was performed on the bright white area indicated in Fig. 8(a-i) (Fig. 8(b)). The results indicated that the atomic percentage of Ni was highest in this region, accompanied by a relatively high proportion of carbon and a small amount of oxygen. This suggests that the bright white particles may represent composite structures with a Ni core, enriched with carbon and oxygen. Furthermore, a detectable amount of Au was found in the particles, indicating that the thin Au layer did not completely prevent the diffusion of Ni atoms during the preceding heat treatment. The exposed Ni particles oxidized and interacted with environmental pollutants, resulting in an interfacial contamination zone. Following Plasma-Ar treatment, significant changes in surface morphology were observed (Fig. 8(c)). The number of bright white particles decreased, and their outlines became more rounded. Additionally, as shown in Fig. 8(c-i), the particle size increased notably, suggesting that argon plasma induced the coarsening of surface particles. In contrast, the sample treated with Plasma-Ar+H₂ (Fig. 8(d) and (d-i)) exhibited smaller surface particles with a more concentrated distribution, demonstrating improved size consistency and spatial uniformity.

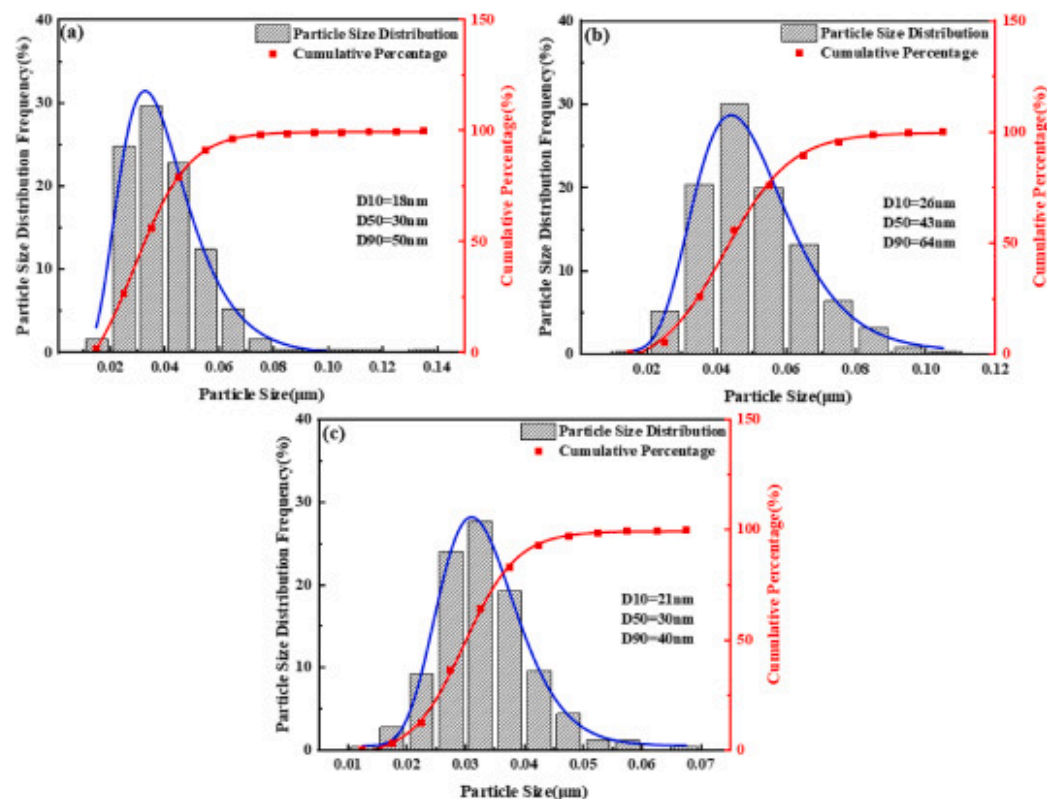


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Fig. 8. Presents SEM images of the BSM chip's plating surface following various surface treatments: (a)-(a-i) No Plasma sample; (b) EDS elemental analysis at position 1 in Fig. (a-i); (c)-(c-i) Plasma-Ar sample; (d)-(d-i) Plasma-Ar+H₂ sample.

The size distribution of surface cellular particles was quantitatively analyzed using image analysis software, with results depicted in Fig. 9. In the No plasma sample, particles displayed a non-uniform distribution, with an average size of 41 ± 5 nm and a broad size range (D10=18 nm, D50=30 nm, D90=50 nm). After Plasma-Ar treatment, the average particle size increased to 50 ± 6 nm (D10=26 nm, D50=43 nm, D90=64 nm). In contrast, following Plasma-Ar+H₂ treatment, the average particle size decreased to 33 ± 4 nm (D10=21 nm, D50=30 nm, D90=40 nm), resulting in a more uniform distribution. These findings indicate that the introduction of H₂ effectively inhibits particle coarsening and enhances the uniformity of the surface morphology.



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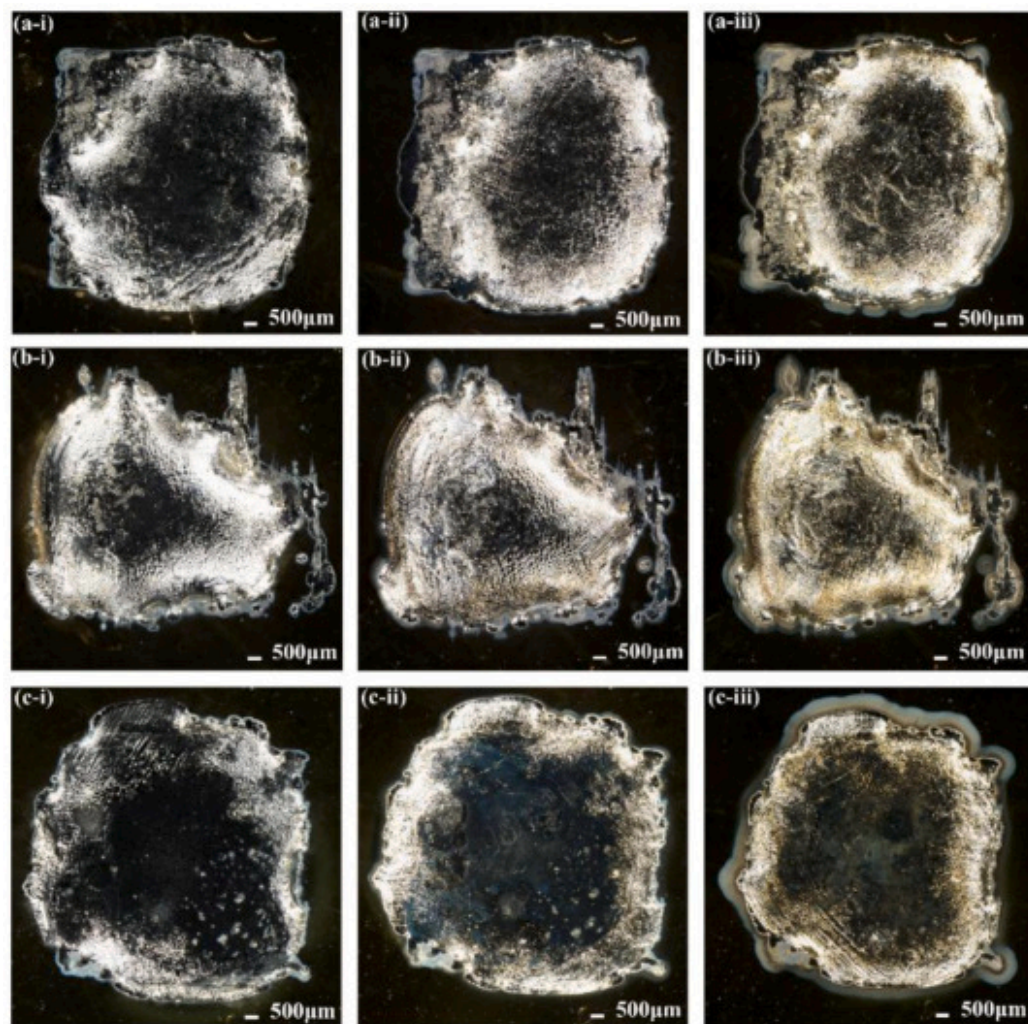
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Fig. 9. Particle size distribution histogram and cumulative frequency curve of the BSM chip coating after various surface treatments: (a) No plasma, (b) Plasma-Ar, (c) Plasma-Ar+H₂.

3.3. Evolution of the spreading ratio

This study systematically investigated the impact of plasma treatment on the wetting reliability of BSM chips during multiple high-temperature reflow processes. It specifically analyzed the alterations in the spreading ratio of indium thermal interface materials on samples subjected to various surface treatments after 1, 3, and 9 reflows using two flux systems: water-soluble and no-clean fluxes.

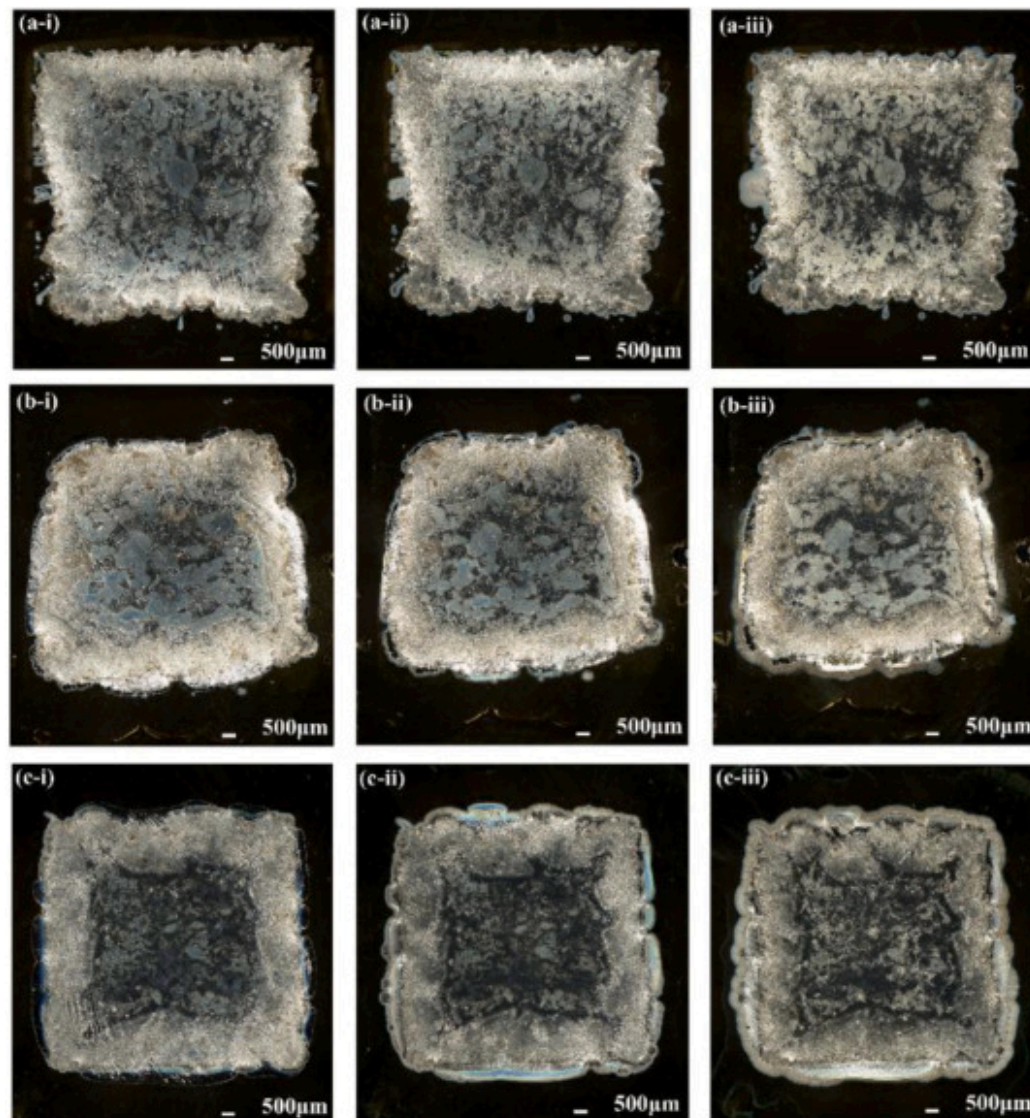
Fig. 10, Fig. 11 present the results, emphasizing the considerable influence of both the surface treatment method and the number of reflow cycles on spreading behavior. An increase in the number of reflows leads to a trend of shrinking across all samples, indicating that the activity of both fluxes progressively declines during the thermal process. As a result, the driving force for interfacial wetting diminishes steadily.



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Fig. 10. Optical micrographs of the spreading morphology of samples with water-soluble flux after different surface treatments: (a-i)–(a-ii) No plasma samples after 1/3/9 high-temperature reflows; (b-i)–(b-ii) Plasma-Ar samples after 1/3/9 high-temperature reflows; (c-i)–(c-ii) Plasma-Ar+H₂ samples after 1/3/9 high-temperature reflows.

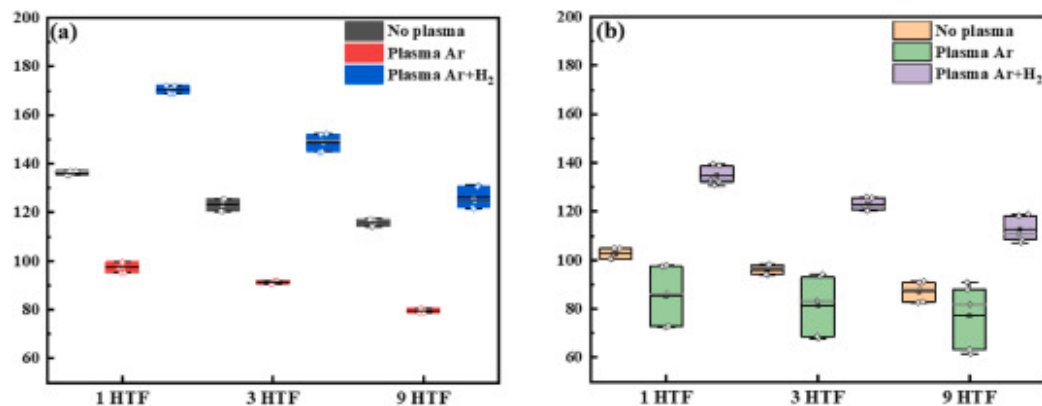


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Fig. 11. Optical micrographs of the spreading morphology of samples with no-clean flux after different surface treatments: (a-i)–(a-ii) No plasma samples after 1/3/9 high-temperature reflows; (b-i)–(b-ii) Plasma-Ar samples after 1/3/9 high-temperature reflows; (c-i)–(c-ii) Plasma-Ar+H₂ samples after 1/3/9 high-temperature reflows.

The spreading ratios of the samples are illustrated in Fig. 12, while the statistical significance results are summarized in Tables S4 and S5. In the water-soluble flux system (Fig. 12(a)), the average spreading ratio for the No plasma samples decreased from 136.0% to 115.7%, indicating a reduction in wetting performance. The Plasma-Ar samples exhibited the lowest wetting performance, with average spreading ratios declining to 97.4%, 91.3%, and 79.6%, respectively, suggesting that the single Ar plasma treatment compromised the interfacial wetting ability. Conversely, the Plasma-Ar+H₂ samples displayed enhanced spreading performance, with average ratios of 170.4%, 148.7%, and 126.1%, maintaining a relatively high level even after nine reflows. A similar trend was observed in the no-clean flux system (Fig. 12(b)). In this case, the average spreading ratio for the No plasma samples fell from 102.6% to 87.0%, while the Plasma-Ar samples decreased from 85.3% to 77.1%. Although the spreading ratio of the Plasma-Ar+H₂ samples also declined (from 134.8% to 112.7%), their performance remained superior, with the ratio after nine reflows still 25.7% higher than that of the No plasma samples.

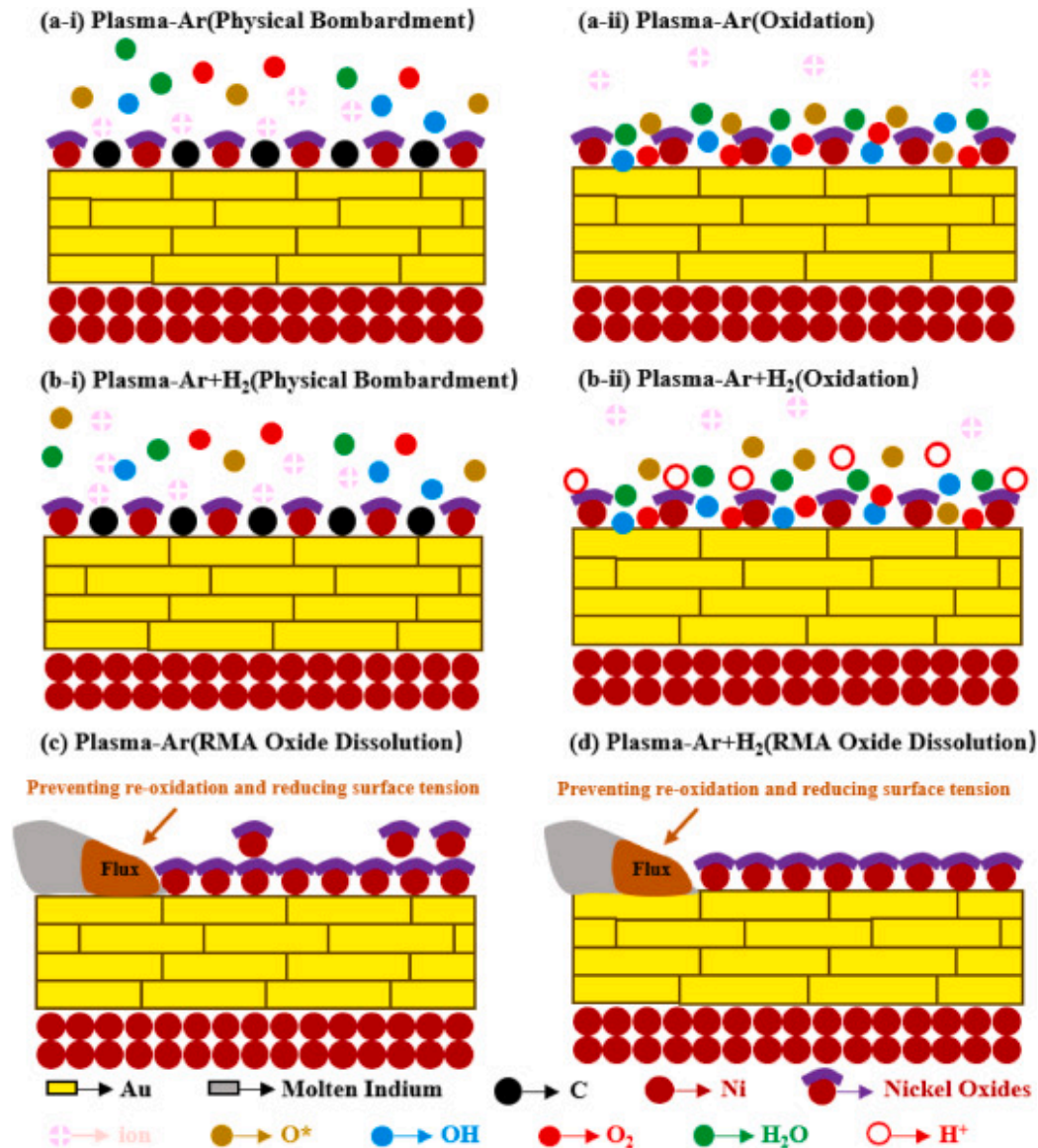


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Fig. 12. Evolution of the spreading ratio of samples with different fluxes after various surface treatments: (a) water-soluble flux samples; (b) no-clean flux samples.

To elucidate the mechanisms underlying these differences, Fig. 13 presents a schematic diagram that illustrates the effects of plasma treatment and the subsequent removal of the oxide layer by flux. In the Plasma-Ar treatment (Fig. 13(a-i, a-ii)), high-energy Ar^+ ions dissociate O_2 and H_2O in the environment while cleaning the surface, generating numerous reactive oxygen radicals ($\cdot\text{O}$, $\cdot\text{OH}$). These radicals oxidize metallic Ni extensively, resulting in the formation of unstable high-valence oxides such as NiOOH and Ni_2O_3 , which leads to a rough oxide layer. In contrast, the Plasma-Ar+ H_2 treatment (Fig. 13(b-i, b-ii)) employs physical bombardment for surface cleaning, while hydrogen radicals ($\cdot\text{H}$), produced by H_2 ionization, serve as potent reducing agents. These radicals convert unstable high-valence oxides into stable low-valence states, such as NiO and $\text{Ni}(\text{OH})_2$, thereby forming a uniform and dense oxide layer. The study by Kumar et al. [44] demonstrated that during the reflow soldering process, the flux is first activated, releasing active components to remove oxides from the coating and solder surfaces, while preventing re-oxidation at high temperatures. Subsequently, the cleaned solder begins to wet the coating surface and gradually spreads, simultaneously propelling the flux forward to continuously promote the wetting process, ultimately forming a complete and reliable solder joint. Traditionally, it has been posited that surface oxidation deteriorates wettability [[45], [46], [47]]. However, this study reveals that in flux-assisted reflow soldering, the characteristics of the oxide layer, rather than its mere presence, are critical for determining spreading performance. Fig. 13(c) demonstrates that the rough and discontinuous island-like oxide layer generated by the Plasma-Ar treatment (Fig. 8(c)) impedes uniform and effective flux removal. Conversely, Fig. 13(d) illustrates that the highly uniform cellular oxide layer produced by the Plasma-Ar+ H_2 treatment (Fig. 8(d)) provides an optimal reaction interface for the flux, facilitating its rapid and even dissolution of the oxides and thereby exposing a clean metal surface that is conducive to solder spreading.



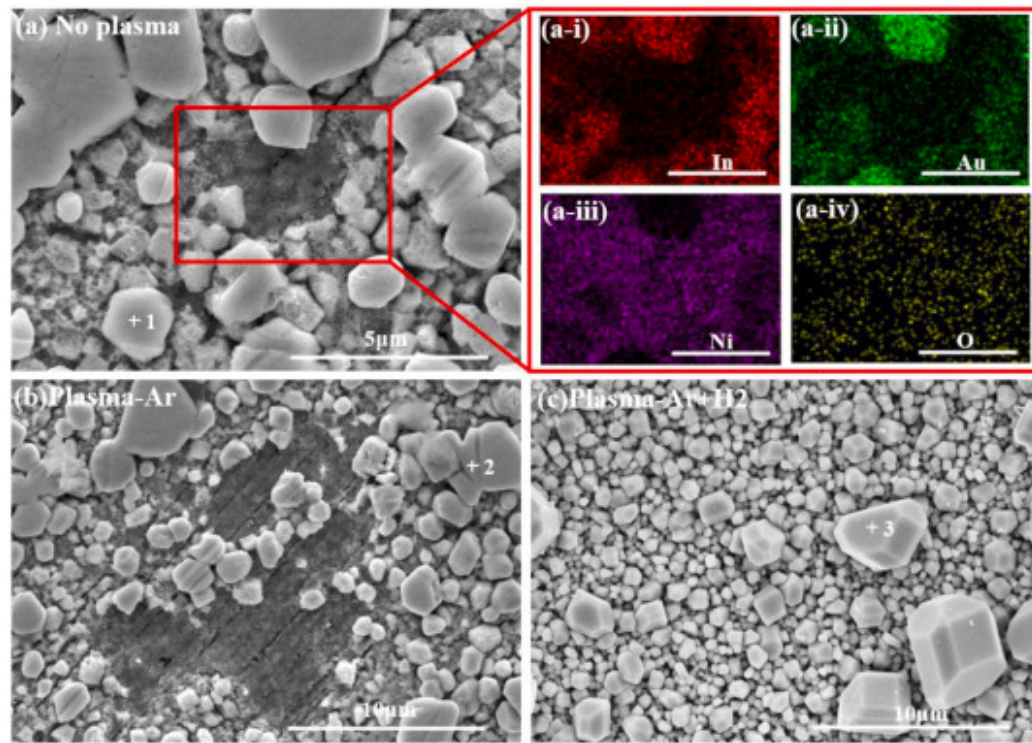
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Fig. 13. Plasma Effects and Flux Action: A schematic of post-treatment coating modification and oxide removal.

3.4. Formation and growth of interfacial intermetallic compound

This study investigates the effects of various plasma treatments on interfacial bonding behavior by etching the indium layer of samples subjected to a single high-temperature reflow. The corrosion of the indium layer allowed for the characterization of the morphology of interfacial IMCs, with results illustrated in [Fig. 14](#). All samples displayed cellular IMCs at their interfaces. To determine the types of IMC phases, EDS was utilized for compositional analysis at various locations, as indicated in [Fig. 14](#). The corresponding compositional statistics are summarized in [Table IV](#). EDS analysis of Points 1 to 3 reveals that the IMC comprises Ni (23.79–25.48at%), In (71.8–74.52at%), and Au (1.69–2.83at%). Phase diagram analysis suggests that the IMC is most likely the (Ni, Au)₂₈In₇₂ phase [[48](#)], [[49](#)], [[50](#)]]. To further verify the phase composition and confirm its homogeneity, EDS line scanning and mapping were conducted on the cross-section of the no plasma sample, as shown in [Figs. S5 to S7](#). The results indicate that the elements In and Ni are uniformly and continuously distributed within the IMC layer, whereas Au is dispersed throughout the IMC. In the No Plasma sample ([Fig. 14\(a\)](#)), the IMC coverage was incomplete, revealing an exposed area in the central region. EDS analysis showed that this exposed region predominantly contained Ni and a minor amount of Au, with no detectable In. After the Plasma-Ar treatment ([Fig. 14\(b\)](#)), the uncovered area at the interfacial IMC expanded further. In contrast, the Plasma-Ar+H₂ sample ([Fig. 14\(c\)](#)) exhibited complete coverage by the IMC, with no exposed areas.



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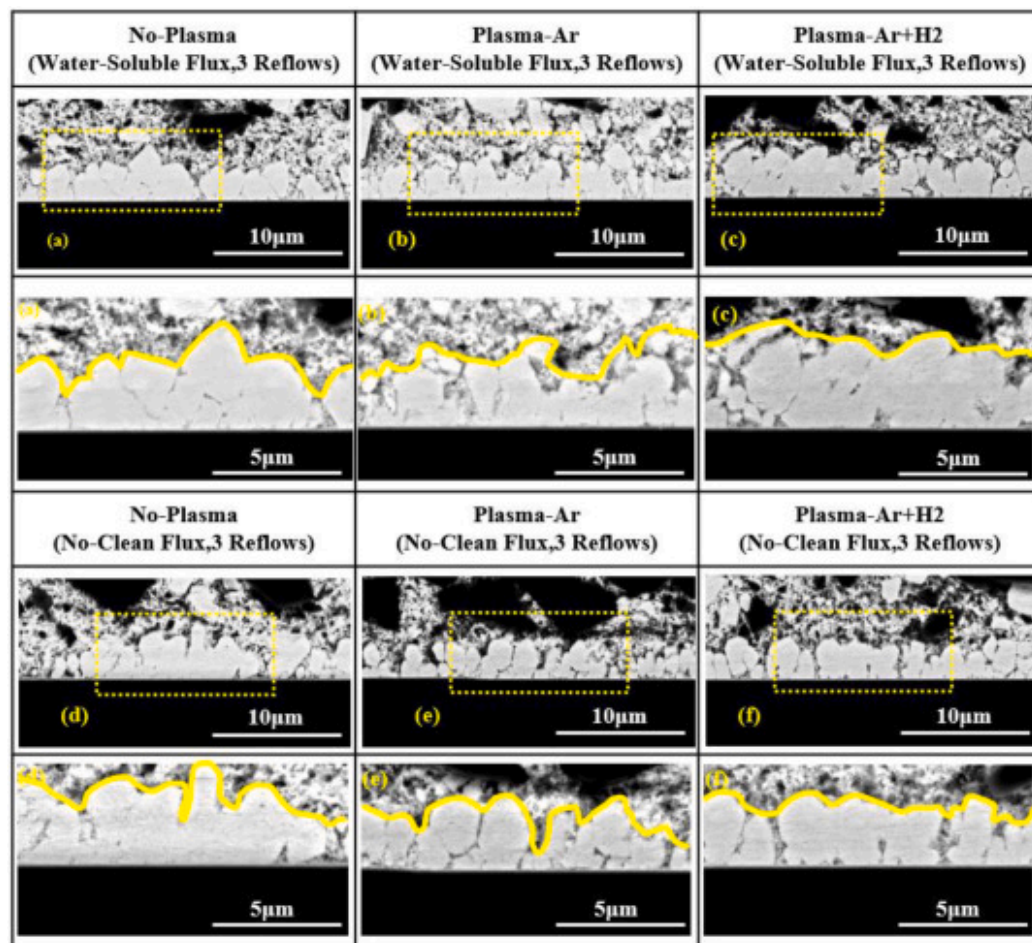
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Fig. 14. Morphology of the IMC at the sample interface after one high-temperature reflow.

Table IV. Analysis of IMC components at the interface of samples treated with different plasmas.

Position	Ni(at%)	In(at%)	Au(at%)	IMC phase
1	23.79	74.52	1.69	$(\text{Ni}, \text{Au})_{28}\text{In}_{72}$
2	24.67	72.5	2.83	
3	25.48	71.8	2.72	

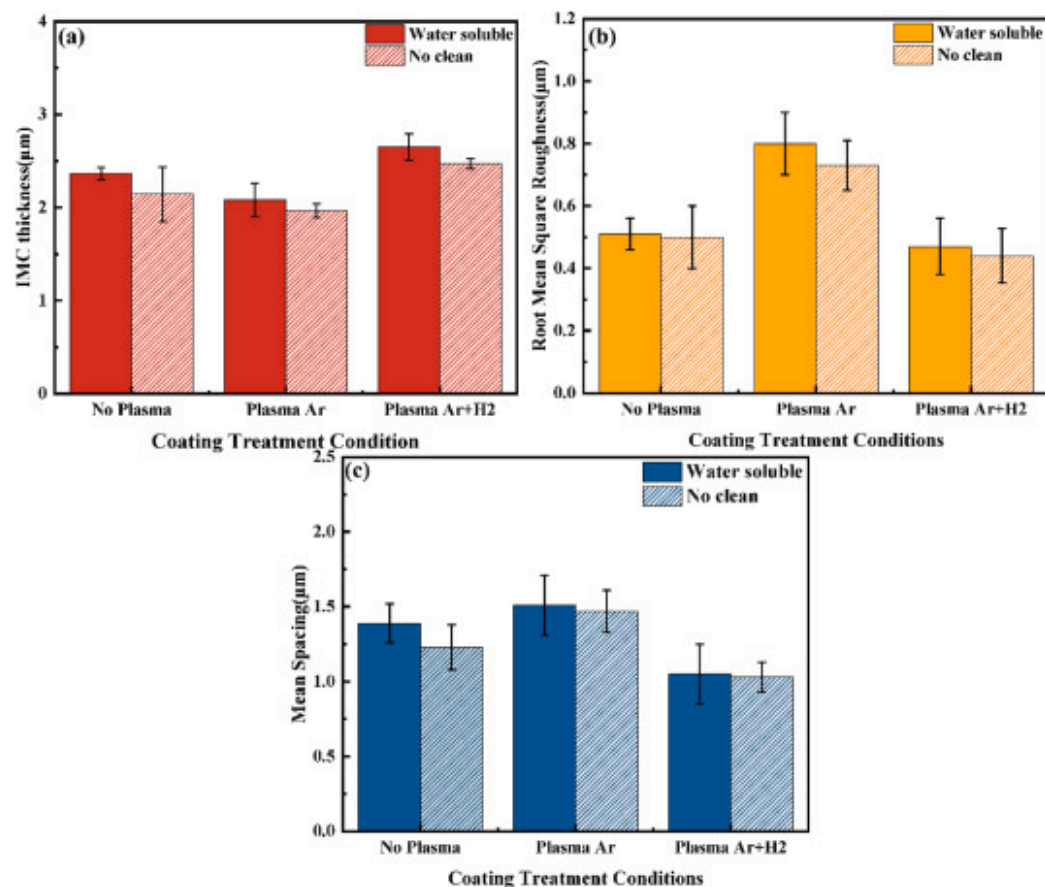
To elucidate the morphological characteristics and growth behavior of the IMC, this study examined the cross-sections of samples following three reflows (Fig. 15) and conducted a statistical analysis of the IMC layer thickness (Fig. 16(a)). In the water-soluble flux system, the average IMC thicknesses for the No plasma, Plasma-Ar, and Plasma-Ar+H₂ samples were 2.36μm, 2.08μm, and 2.65μm, respectively. In the no-clean flux system, the corresponding thicknesses were 2.14μm, 1.96μm, and 2.47μm, respectively. In both systems, the Plasma-Ar sample exhibited the thinnest IMC layer, while the Plasma-Ar+H₂ sample had the thickest. Notably, the overall IMC thickness in samples utilizing the no-clean flux was approximately 10% lower, primarily due to differences in chemical activity and interfacial cleaning efficiency between the no-clean and water-soluble fluxes. The earlier analysis indicates that the non-uniform oxide layer produced by Plasma-Ar treatment obstructs solder spreading and limits the interfacial reaction area, thereby inhibiting IMC growth. Conversely, the uniform oxide layer generated by Plasma-Ar+H₂ treatment is more readily removed by the flux, creating optimal conditions for the interfacial reaction. Consequently, this promotes significant IMC formation and growth.



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Fig. 15. Cross-sectional microstructure of the sample after three high-temperature reflows.



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Fig. 16. Statistics of IMC characteristic parameters of the sample after three high-temperature reflows: (a) Average thickness, (b) Root mean square roughness, (c) Mean spacing.

To quantitatively characterize the interfacial microstructural features formed between indium thermal interface materials and BSM chips with different surface treatments, this study adopts the root mean square roughness (R_{rms}) and the average spacing of the interfacial profile (λ_{ave}) as key evaluation parameters. Specifically, R_{rms} is defined as the root mean square value of the profile

deviations within the sampling length, and its calculation formula is given as [14,51]:

$$R_{rms} = \sqrt{\frac{1}{N} \sum_{i=1}^N y_i^2} \quad (1-2)$$

λ_{ave} is the average value of the profile unevenness spacing λ_i within the sampling length, and its formula is [52,53]:

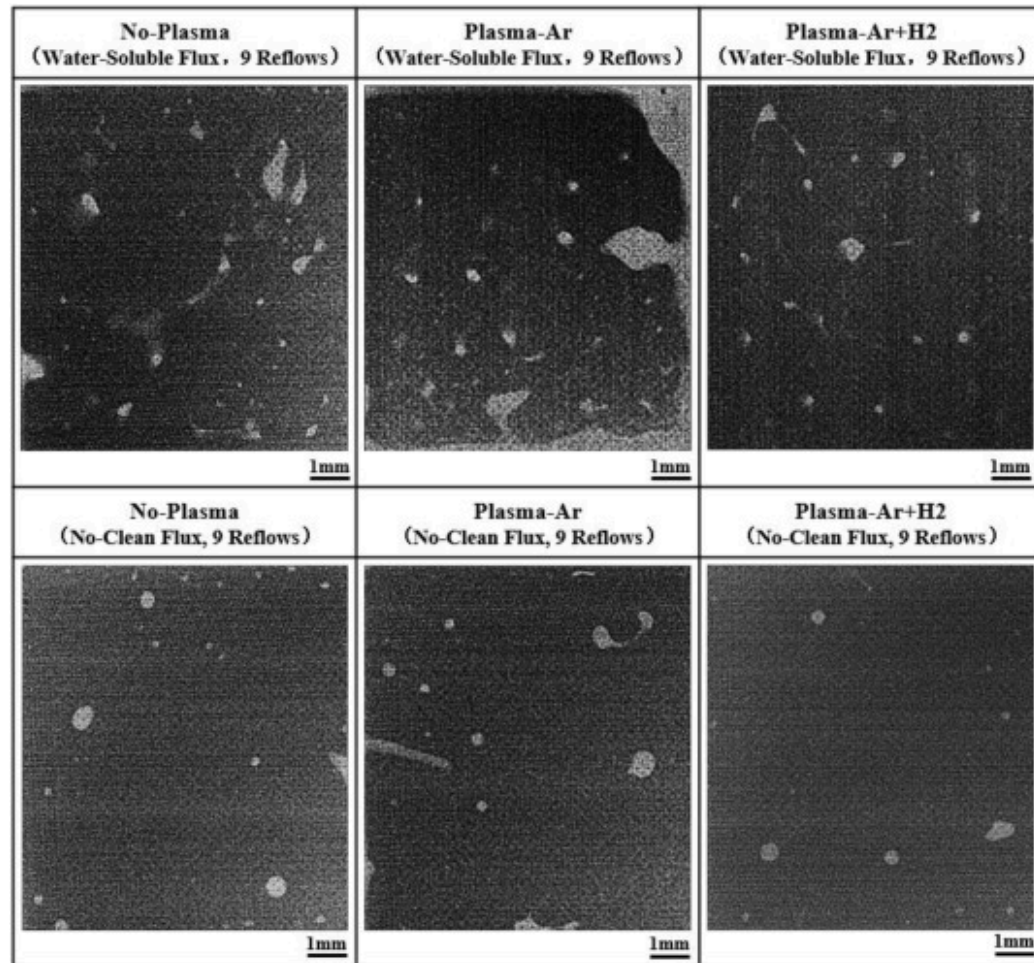
$$\lambda_{ave} = \frac{1}{N} \sum_{i=1}^N \lambda_i \quad (1-3)$$

Fig. 16(b) and (c) demonstrate that in the water-soluble flux system, the R_{rms} values for the interfacial IMCs of the No plasma, Plasma-Ar, and Plasma-Ar+H₂ samples are 0.51 μm , 0.80 μm , and 0.47 μm , respectively. The corresponding λ_{ave} values for these samples are 1.39 μm , 1.51 μm , and 1.05 μm . A similar trend is evident in the no-clean flux system, where the R_{rms} values are 0.50 μm , 0.73 μm , and 0.44 μm , and the λ_{ave} values are 1.23 μm , 1.47 μm , and 1.03 μm , respectively. These results indicate that the Plasma-Ar+H₂ treatment yields the flattest and densest IMC morphology.

Existing studies indicate that the roughness of IMC directly influences wettability [11]. Rough surfaces can anchor the three-phase contact lines through a "pinning effect," which increases the energy barrier that the wetting front must overcome to advance. This phenomenon results in an elevated contact angle and a reduced diffusion area of the solder. Conversely, the flat IMC morphology produced after Plasma-Ar+H₂ treatment significantly lowers this energy barrier and effectively mitigates the pinning effect, thereby promoting the spreading of solder.

3.5. Effect of sample void size percentage on thermal performance

This study employed X-ray imaging to conduct a non-destructive analysis of the indium layer coverage on samples subjected to nine reflow treatments, with results illustrated in Fig. 17. Within the water-soluble flux system, plasma treatments notably influenced the indium layer coverage: the Plasma-Ar sample demonstrated significant indium loss at the corners, the No plasma sample exhibited only minimal corner loss, and the Plasma-Ar-H₂ sample achieved complete and uniform coverage. A comparable trend was noted in the no-clean flux system.



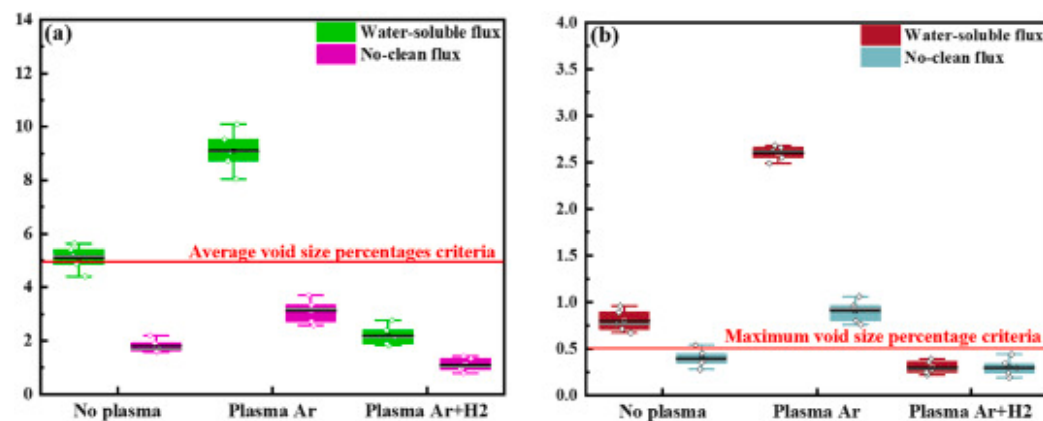
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Fig. 17. X-Ray image of the sample chip area after 9 high-temperature reflows.

As illustrated in Fig. 18, a quantitative statistical analysis was performed on the void size percentage across all samples, with the significance test results detailed in Table S4 and Table S5. In the water-soluble flux system, the average void size percentages for the No Plasma, Plasma-Ar, and Plasma-Ar+H₂ samples were 5.1%, 9.1%, and 2.2%, respectively. In the no-clean flux system, the corresponding values were 1.8%, 3.1%, and 1.1%. Although the chemical activity of the fluxes resulted in numerical variations,

both systems displayed a consistent trend: samples treated with Plasma-Ar+H₂ exhibited the lowest void size percentage, while those treated with Plasma-Ar showed the most pronounced void defects. High-reliability electronic products impose stringent requirements on thermal and mechanical performance, typically necessitating a reduction in the average solder void size percentage from 10 to 15% to below 5% [[54], [55], [56], [57]]. To this end, we established an internal acceptance criterion that is more stringent than general standards: an average void size percentage below 5% and a maximum single void size percentage below 0.5%. In this study, both the No Plasma and Plasma-Ar samples in the water-soluble system failed to meet these criteria. In the no-clean system, although the No Plasma sample met the requirement for average void size percentage, its maximum void size percentage still exceeded the specified limit. In contrast, following treatment with the Plasma-Ar+H₂ process, the average void size percentages for the two flux systems were 2.2% and 1.1%, respectively, with maximum void size percentages of 0.3% and 0.2%—all exceeding the internal acceptance standards.



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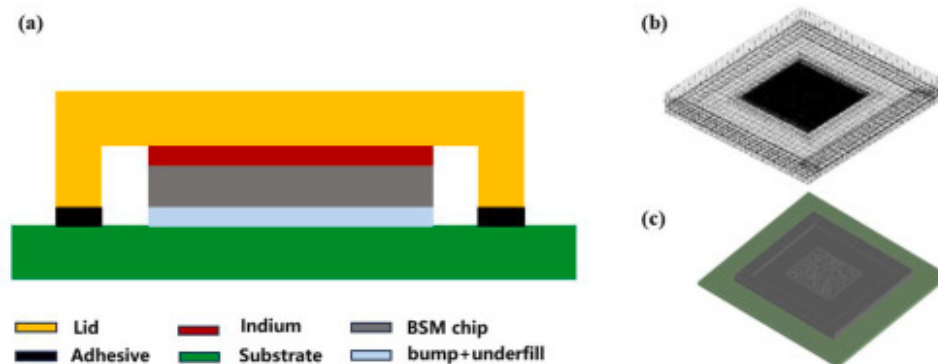
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Fig. 18. Void size percentage of samples with different fluxes after various surface treatments: (a) average void size percentages; (b) maximum void size percentage.

The phenomena described can be effectively elucidated by the mechanism through which surface states influence wettability. A uniform oxide layer forms on the surface of the coating treated with Plasma-Ar+H₂, thereby optimizing the flux function. This uniformity promotes even spreading of the indium solder and effectively suppresses void formation due to inadequate wetting. In contrast, contaminants and an oxide layer on the surface of the No plasma sample result in poor interface contact and

localized defects. Although Plasma-Ar treatment enhances surface cleanliness, the non-uniform oxide layer it generates exacerbates overall wettability, leading to the most significant void defects. In summary, the initial surface cleanliness and the uniformity of the oxide layer on the coating jointly determine the wetting behavior of the indium thermal interface material, serving as critical factors that influence its final coverage and bonding reliability.

This study further examines the impact of void evolution on the thermal performance of the package, based on variations in void size percentage across samples subjected to different surface treatments. A three-dimensional model of the indium layer, incorporating void structures, was constructed through grayscale processing of X-ray images and subsequent mapping into SolidWorks modeling software. To assess the influence of void distribution on the temperature field, geometric modeling and mesh generation of the FCBGA package structure were performed, as illustrated in Fig. 19, with local mesh refinement applied to the indium layer. The model dimensions and material parameters are detailed in Table V.



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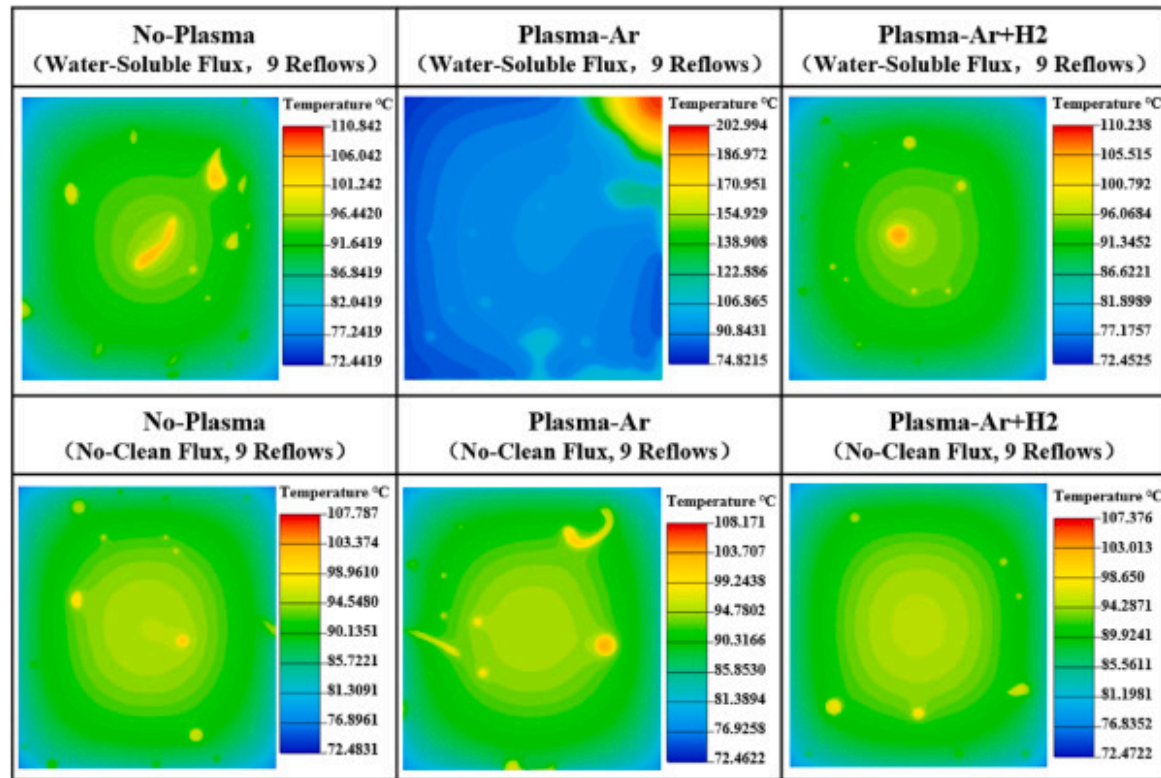
Fig. 19. (a) Cross-sectional material layers of the package; (b) Meshed computational model; (c) 3D structural representation of the package.

Table 5. Dimensions and material parameters of the finite element model.

Material	Length (mm)	Width (mm)	Thickness (mm)	Thermal Conductivity (W/(m·K))
Lid	46.7	46.7	3	385

Material	Length (mm)	Width (mm)	Thickness (mm)	Thermal Conductivity (W/(m·K))
Indium	22	21	0.23	82
BSM chip	22	21	0.75	117.5
Bump+underfill	22	21	0.75	58(Micro bump); 0.52(Underfill)
Adhesive	3.78	3.78	0.23	0.2
Substrate	50	50	1.542	0.579(CORE); 385(Copper)

This section details steady-state thermal simulations performed with Ansys Icepak (2023 R2) software to examine the effect of void size percentage within the indium layer on the thermal dissipation performance of the chip. A power dissipation of 1000W was applied to the chip, while a water-cooling system was configured at the system end to sustain a constant cold plate temperature of 60°C. Fig. 20 illustrates the temperature distributions at the chip–indium interface for samples subjected to various surface treatments.

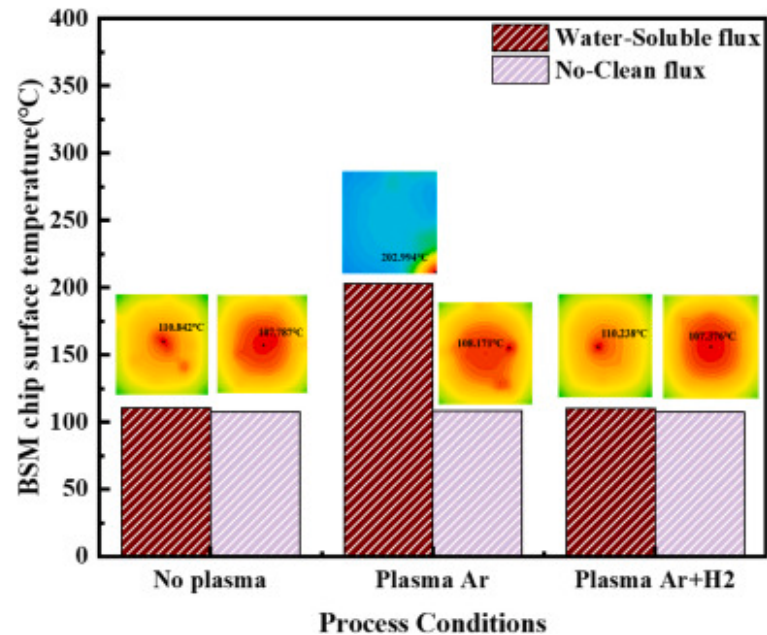


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Fig. 20. Temperature field of the sample chip area after 9 high-temperature reflows.

Fig. 21 illustrates the temperature variations on the chip surface of samples subjected to different plasma treatments under both water-soluble and no-clean flux systems. Simulation results consistently indicate that void defects significantly impair the thermal performance of the package, with a positive correlation observed between the void size percentage and the chip surface temperature. In the water-soluble flux system, the Plasma-Ar sample, which exhibited the highest void size percentage, reached a peak surface temperature of 202.994°C. In contrast, samples with lower void size percentages specifically, the No Plasma and Plasma-Ar+H₂ samples recorded maximum temperatures of 110.842°C and 110.238°C, respectively. This represents a reduction of more than 45% compared to the Plasma-Ar sample, highlighting a substantial improvement in heat dissipation performance.



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Fig. 21. Surface temperature of the sample chip area after 9 high-temperature reflows.

In summary, irrespective of the flux system, the Plasma-Ar+H₂ treatment consistently achieved optimal thermal performance by producing the lowest percentage of void size. In contrast, the Plasma-Ar sample demonstrated the poorest heat dissipation, attributable to its significant void defects. This finding thermally substantiates the essential role of minimizing void size percentage in improving the thermal performance of the samples.

4. Conclusions

This study systematically examines the impact of plasma treatment on the surface properties of BSM chip coatings and their bonding reliability with indium thermal interface materials. The findings reveal that Ar/H₂ plasma treatment significantly enhances wettability during high-temperature reflow. XPS analysis shows that while pure Ar plasma effectively cleans the surface, it also promotes nickel oxidation, resulting in an increased formation of Ni³⁺ species. In contrast, the inclusion of H₂

provides a reducing effect, effectively mitigating excessive oxidation and facilitating the development of an oxide layer primarily composed of Ni^{2+} . In terms of surface properties, the sample without plasma treatment exhibited a surface energy of 28.25 mN/m and an average particle size of 41 ± 5 nm. Ar plasma treatment elevated the surface energy to 61.00 mN/m but also caused particle coarsening, increasing the average size to 50 ± 6 nm. Conversely, the Ar/ H_2 plasma treatment synergistically enhanced both properties: the surface energy increased to 72.50 mN/m, while the particle size was refined to 33 ± 4 nm, confirming that the addition of hydrogen effectively inhibits coarsening and improves surface activation. After nine high-temperature reflows, the Ar/ H_2 -treated sample exhibited optimal interfacial performance, achieving a spreading ratio of 112.7%, the flattest IMC layer with a root mean square roughness (R_{rms}) of $0.44 \mu\text{m}$, and the lowest void size percentage at 1.1%. In contrast, the untreated sample demonstrated a spreading ratio of 87.0%, an R_{rms} of $0.50 \mu\text{m}$, and a void size percentage of 1.8%. The Plasma-Ar sample, negatively impacted by a non-uniform oxide layer, displayed further degraded performance, with a spreading ratio of 77.1%, an R_{rms} of $0.73 \mu\text{m}$, and a void size percentage as high as 3.1%. Thermal simulation results established a clear correlation between void size percentage and chip operating temperature. In the water-soluble flux system, the Plasma-Ar sample, which had the highest void size percentage, reached a peak temperature of 202.994°C . In contrast, the untreated and Ar/ H_2 -treated samples recorded significantly lower temperatures of 110.842°C and 110.238°C , respectively, indicating a temperature reduction of over 45%. A similar trend was observed in the no-clean flux system: the Ar-treated sample recorded the highest temperature at 108.171°C , followed by the untreated sample at 107.787°C , and the Ar/ H_2 sample at the lowest temperature of 107.376°C . These results further validate the efficacy of Ar/ H_2 plasma treatment in enhancing overall package thermal performance.

CRedit authorship contribution statement

Yiou Qiu: Writing – review & editing, Writing – original draft, Software, Methodology, Conceptualization. **Zhen Liu:** Software, Data curation. **Linzheng Fu:** Visualization, Investigation. **Ping Wu:** Validation, Software. **Liu Li:** Validation, Software. **Danni Cao:** Validation, Software. **Wenjuan Guo:** Validation, Software. **Oufei Liu:** Validation, Software. **Shangde Xu:** Validation, Software. **Xia Feng:** Validation, Software. **Minxuan Liu:** Supervision. **Xiaodong Teng:** Supervision. **Wenhui Zhu:** Writing – review & editing, Funding acquisition. **Liancheng Wang:** Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

The following is the Supplementary data to this article.

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Multimedia component 1.

[Recommended articles](#)

Data availability

Data will be made available on request.

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